Lecture 5: detailed solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to `Listas/lista-reais-lecture-05.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 5 — Repeated games and bargaining

Item 1 (item 41 of the combined list)**: Cabral Exercise 7.13** Foundations · difficulty 1/5

Source: Cabral Exercise 7.13, page 182. The statement is reproduced verbatim in the list (item 41); it is not repeated here.

(a) What is asked. Prove that when Cabral's Prisoner's Dilemma (Figure 7.1) is repeated a known number T of times, the only subgame-perfect equilibrium (SPE) has (B, R) played in every period.

(b) Method. Backward induction on the finitely repeated game $G(T, \delta)$ (Lecture 4, slide 22: each period starts a new subgame; slide 17: an SPE is a Nash equilibrium in every subgame; slides 9–10: backward induction). The result is Tadelis Prop. 10.1 (§10.1, p. 192), a special case of Prop. 9.3 (§9.4, p. 181): if the stage game has a unique Nash equilibrium, the finitely repeated game has a unique SPE, the stage equilibrium in every period after every history. Cabral sets up the repeated Prisoner's Dilemma in §7.3 (pp. 172–174). The proof is an induction on the number of periods left.

(c) Solution. Notation. Periods $t = 1, \ldots, T$; h^{t-1} is the history of action profiles before period t; player i's payoff is $\sum_{t=1}^{T} \delta^{t-1} u_i(a^t)$ with $\delta \in (0, 1]$ (slide 22). Stage payoffs (Figure 7.1, player 1 first):

	L	R
T	5, 5	3, $\underline{6}$
B	$\underline{6}$, 3	$\underline{4}$, $\underline{4}$

Step 1: the stage game. For player 1, B beats T against both columns: $6 > 5$ against L and $4 > 3$ against R. So B is strictly dominant; by symmetry R is strictly dominant for player 2. The stage game therefore has exactly one Nash equilibrium, (B, R) with payoffs $(4, 4)$, and no mixed equilibrium (a strictly dominated action gets zero weight in any equilibrium).

Step 2: the last period, $t = T$. Take any history h^{T-1}. The payoffs already earned, $\sum_{t<T} \delta^{t-1} u_i(a^t)$, are fixed and nothing follows period T. So the subgame after h^{T-1} is the stage game with every payoff multiplied by $\delta^{T-1} > 0$ and a constant added. Neither operation changes best replies, so the only Nash equilibrium of that subgame is (B, R). An SPE must therefore play (B, R) in period T after *every* history.

Step 3: the induction step. Suppose that in every SPE, (B, R) is played in periods $k+1, \ldots, T$ after every history. Take any history h^{k-1}. Whatever is played in period k, the continuation from $k+1$ on is (B, R) in each of the $T - k$ remaining periods, worth

$$W \;=\; \sum_{t=k+1}^{T} \delta^{t-1} \cdot 4,$$

the same constant for both players and independent of the period-k action. Period k's subgame therefore reduces to the stage game scaled by δ^{k-1} plus the constant W (plus the sunk past payoffs). Its unique Nash equilibrium is again (B, R). The claim holds for period k after every history.

Step 4: conclusion. By induction from $k = T$ down to $k = 1$, the unique SPE is: each player plays B (player 1) and R (player 2) in every period after every history. Its payoff is $4 \sum_{t=1}^{T} \delta^{t-1}$

for each player. The argument uses neither the value of δ nor the value of T, only that T is finite and known.

One-stage deviation check. At any history in period k, a single deviation to T (or L) changes only the period-k payoff, from 4 to 3 if the opponent plays R, while the continuation stays (B, R). The deviation loses δ^{k-1}, so no one-stage deviation pays (Tadelis Thm. 9.1, p. 185).

A stronger statement (Nash equilibrium paths). Even without subgame perfection, every Nash equilibrium of the repeated game has the *path* (B, R) in every period. Player 1 can always play B, which gives him at least 4 per period (row B pays 6 or 4), and 4 is also what (B, R) pays. Going backwards along the path: if the path had T (or L) in some period, the player could switch to B (or R) there and in every later period, earn one more in that period, and get at least 4 later, which is what the path gives later by the induction. The punishment threats that could deter this are worth nothing, because the minimax payoff, 4, equals the equilibrium payoff. `check_C_L05.py` confirms this for $T = 2$ by brute force.

(d) Interpretation. Cooperation in a finitely repeated game needs a threat for the future, and here there is none. In the last period there is no future, so the players defect. In the second-to-last period the last period's outcome is already fixed, so there is again nothing to protect, and the argument unravels back to period 1. Cabral's point in §7.3 is that an *indefinitely* repeated game, one that ends after each period with probability $1 - \alpha$ (Cabral's notation, p. 172) and so has no known last period, is needed for retaliation to sustain (T, L). A known end date, however distant, removes it.

(e) Verification. `check_C_L05.py` checks that B and R are strictly dominant and that (B, R) is the unique Nash equilibrium of the stage game and of the folded game with any added constant. For $T = 2$ it enumerates all $32 \times 32 = 1024$ pure strategy profiles (a period-1 action and a map from the four period-1 outcomes to a period-2 action) at $\delta \in \{\frac{1}{10}, \frac{1}{2}, 1\}$. It finds exactly one SPE (always B, always R), and every Nash equilibrium has the path $(B, R), (B, R)$. Output: "Cabral 7.13: unique stage NE (B,R); T=2 brute force (1024 profiles x 3 deltas): 1 SPE, all NE paths (B,R),(B,R)". Cabral prints no answer; the result matches Tadelis Prop. 10.1.

(f) Common mistakes.

- Showing only the outcome, "(B, R) every period", and not the strategy "(B, R) after *every* history". An SPE prescribes play off the path too, and the induction must cover every history.
- Arguing that δ large enough or T long enough allows cooperation. With a known last period and a unique stage equilibrium, neither matters.
- Folding the game forward ("if they cooperate today they will cooperate tomorrow") instead of starting at period T. Backward induction runs from the end.

Answer. B and R are strictly dominant, so (B, R) is the unique stage equilibrium. In period T every subgame is the stage game, so (B, R) is played after every history; given that, the continuation from any period k on does not depend on period-k actions, so period k is again the stage game plus a constant and (B, R) is played. By induction the unique SPE plays (B, R) in every period after every history, for any T and any δ.

Item 2 (item 42 of the combined list)**: Tadelis Exercise 10.2** Foundations · difficulty 2/5

Source: Tadelis Exercise 10.2, page 215. The statement is reproduced verbatim in the list (item 42); it is not repeated here.

(a) What is asked. In the infinitely repeated 3×3 Prisoner's Dilemma variant, find the discount factors for which (M, C) in every period, and then (T, L) in every period, can be sustained, and explain why the two answers differ.

(b) Method. Grim-trigger strategies and the one-stage deviation principle (Tadelis §10.3, pp. 197–200: grim trigger p. 198, Prop. 10.3 p. 198, condition (10.3) p. 200), and the general threshold (10.5), p. 202:

$$\delta \;\geq\; \delta^* = \frac{v_i^d - v_i^c}{v_i^d - v_i^*},$$

with v_i^c the cooperative stage payoff, v_i^d the best one-period deviation payoff, and v_i^* the stage Nash payoff used as punishment (`Notas/L05/L05-05`). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17.

(c) Solution. Notation. Present value $v_i = \sum_{t\geq1}\delta^{t-1}u_i(a^t)$ (Tadelis Def. 10.2, p. 193). Best replies are underlined:

	L	C	R
T	$6,\ 6$	$-1,\ 7$	$-2,\ \underline{8}$
M	$7,\ -1$	$4,\ 4$	$-1,\ \underline{5}$
B	$\underline{8},\ -2$	$\underline{5},\ -1$	$\underline{0},\ \underline{0}$

Step 1: the stage equilibrium. Row B beats T ($8>6$, $5>-1$, $0>-2$) and M ($8>7$, $5>4$, $0>-1$) against every column, so B is strictly dominant for player 1. The game is symmetric, so R is strictly dominant for player 2. The unique stage Nash equilibrium is (B, R), with $v_i^* = 0$. It is also each player's minimax payoff: player 2 can hold player 1 to $\min_c \max_r u_1(r, c) = \min\{8, 5, 0\} = 0$ by playing R, and B guarantees player 1 at least $\min\{8, 5, 0\} = 0$.

Step 2: the strategies. For a target profile (a_1, a_2), grim trigger is:

- *cooperation phase*: in period 1, and after any history in which (a_1, a_2) was played in every period, player 1 plays a_1 and player 2 plays a_2;
- *punishment phase*: after any other history, player 1 plays B and player 2 plays R forever.

By Prop. 10.3 this is an SPE if and only if no player gains from a single deviation at a single history. There are two kinds of histories, so there are two checks.

Step 3: punishment phase (both parts). The continuation is (B, R) forever whatever is played today, so a one-period deviation changes only today's payoff. Against R, player 1 gets 0 with B, -1 with M and -2 with T. Deviating loses, for every δ, and player 2 is symmetric.

(a) (M, C)**.** Cooperation phase. Staying gives 4 every period:

$$v_1^{\text{coop}} = \frac{4}{1-\delta}.$$

The best one-period deviation against C is B ($5>4>-1$), and it starts the punishment phase, which pays 0 forever:

$$v_1^{\text{dev}} = 5 + \delta\cdot\frac{0}{1-\delta} = 5.$$

No deviation pays iff

$$\frac{4}{1-\delta} \geq 5 \iff 4 \geq 5-5\delta \iff 5\delta \geq 1 \iff \delta \geq \frac{1}{5}.$$

Check with (10.5): $\delta^* = \frac{5-4}{5-0} = \frac{1}{5}$. Player 2's problem is the same by symmetry. So (M, C) in every period is an SPE outcome iff $\delta \geq \frac{1}{5}$.

(b) (T, L)**.** Staying gives $\frac{6}{1-\delta}$. Against L, player 1's best deviation is B (payoff 8), not M (payoff 7). It is followed by 0 forever:

$$\frac{6}{1-\delta} \geq 8 \iff 6 \geq 8-8\delta \iff 8\delta \geq 2 \iff \delta \geq \frac{1}{4}.$$

Check: $\delta^* = \frac{8-6}{8-0} = \frac{1}{4}$. The same holds for player 2. So (T, L) in every period is an SPE outcome iff $\delta \geq \frac{1}{4}$.

These thresholds cannot be beaten. The punishment (B, R) forever holds the deviator to 0, his minimax payoff, and he can guarantee 0 in every period by playing B. So in *any* equilibrium (any punishment, even a Nash equilibrium that is not subgame perfect) a deviation from the stationary path is worth at least 5 in (a) and 8 in (b). The thresholds $\frac{1}{5}$ and $\frac{1}{4}$ are therefore necessary as well as sufficient.

Why the answers differ. Write (10.5) as $\delta^* = g/(g+\ell)$, with $g = v^d - v^c$ the one-period gain and $\ell = v^c - v^*$ the per-period loss once punished:

target	g	ℓ	g/ℓ	δ^*
(M, C)	$5-4=1$	$4-0=4$	1/4	1/5
(T, L)	$8-6=2$	$6-0=6$	1/3	1/4

Moving from (M, C) to (T, L) raises the value of cooperating (the loss ℓ rises from 4 to 6), but it raises the temptation more in proportion (the gain g doubles, from 1 to 2). The ratio g/ℓ rises, so more patience is needed. For $\delta \in [\frac{1}{5}, \frac{1}{4})$ the moderate outcome (M, C) can be sustained and the efficient (T, L) cannot.

(d) Interpretation. More ambitious cooperation is harder to sustain when it also creates a bigger temptation. (T, L) is better for both players than (M, C), but at (T, L) each player gives up a lot today by not grabbing the 8. Between moderately patient partners the self-enforcing agreement is the modest one. The same trade-off runs through collusion (Tadelis §10.4): higher collusive prices raise both the gain from undercutting and the value of the cartel.

(e) Verification. `check_C_L05.py` confirms that B and R are strictly dominant, that (B, R) is the unique stage equilibrium with payoff 0 = minimax, and that the best deviations are B against C (5) and B against L (8, versus 7 for M). Using exact fractions it tests the cooperation-phase inequality at δ^*, $\delta^* \pm \frac{1}{1000}$ and on the grid $\delta = k/1000$: it holds exactly for $\delta \geq \frac{1}{5}$ in (a) and $\delta \geq \frac{1}{4}$ in (b). The punishment phase has no profitable deviation. Output: "Tadelis 10.2: stage NE (B,R) (0,0); (a) delta* = 1/5, (b) delta* = 1/4; grid k/1000 and +-1/1000 around each cutoff pass". Tadelis prints no answer.

(f) Common mistakes.

- Using M as the deviation from (T, L), because it is "next door". That gives $\frac{7-6}{7-0} = \frac{1}{7}$, which is wrong: the deviation to use is the *best* reply, B.

- Checking only the cooperation phase. Prop. 10.3 needs the punishment phase too; here it holds because (B, R) is a stage equilibrium, and that must be said.
- Reading "(T, L) is better for both" as "(T, L) is easier to sustain". The threshold depends on the ratio of temptation to loss, not on how good the outcome is.

Answer. Unique stage equilibrium (B, R), payoff $(0, 0)$ (also the minimax). With grim trigger (revert to (B, R) forever): (a) (M, C) is sustained iff $\frac{4}{1-\delta} \geq 5$, i.e. $\delta \geq \frac{1}{5}$; (b) (T, L) is sustained iff $\frac{6}{1-\delta} \geq 8$, i.e. $\delta \geq \frac{1}{4}$. Since the punishment is the minimax, no other strategies do better. The answers differ because $\delta^* = g/(g + \ell)$: at (T, L) the temptation doubles ($g = 2$ against 1) while the loss from punishment rises only from 4 to 6.

Item 3 (item 43 of the combined list)**: Muñoz-Garcia Ch. 6 Exercise 1** Foundations · difficulty 2/5

Source: Muñoz-Garcia Ch. 6 Exercise 1, page 274. The statement is reproduced verbatim in the list (item 43); it is not repeated here.

(a) What is asked. In the infinitely repeated Prisoner's Dilemma of Table 6.1, can (NC, NC) be an SPE outcome when a deviation is punished by (C, C) for (a) one period, (b) two periods, and (c) for ten periods, what is the threshold δ?

(b) Method. Temporary Nash reversion ("length-k punishment") and the one-stage deviation principle (Tadelis Prop. 10.3, p. 198; grim trigger and (10.3), pp. 198–200; the general threshold (10.5), p. 202). Compare the stream from cooperating with the stream from deviating once and then being punished for k periods; the two streams coincide from period $k+2$ on. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md`.

(c) Solution. Notation. Stage payoffs (player 1 first): $(C,C) = (2,2)$, $(C,NC) = (6,0)$, $(NC,C) = (0,6)$, $(NC,NC) = (3,3)$. Present values $\sum_{t\geq 0} \delta^t u_i^t$, measured from the period in which a player considers deviating. A length-k punishment strategy has two phases:

- *cooperation*: play NC in period 1 and as long as no deviation from the prescribed actions has occurred;
- *punishment*: after a deviation, play C for k periods, then return to cooperation (a deviation during the punishment restarts it).

Step 1: the stage game. C is strictly dominant ($2 > 0$ against C, $6 > 3$ against NC), so (C,C), with $(2,2)$, is the unique stage equilibrium. The deviation gain is $g = 6 - 3 = 3$; each punishment period costs $\ell = 3 - 2 = 1$.

Step 2: the general condition. Cooperating gives $3 + 3\delta + \dots$. Deviating gives 6 now, 2 in each of the next k periods, then 3 forever. From period $k+1$ on (counting the current period as 0) the streams are equal, so the comparison is over periods $0, \dots, k$:

$$3 + 3\sum_{j=1}^{k} \delta^j \;\geq\; 6 + 2\sum_{j=1}^{k} \delta^j \iff \sum_{j=1}^{k} \delta^j \;\geq\; 3 \iff \frac{\delta(1-\delta^k)}{1-\delta} \;\geq\; 3.$$

In words: the one-period gain, 3, must be covered by k discounted losses of 1 each. Since $\sum_{j=1}^{k} \delta^j < k$ for $\delta < 1$, no punishment of length $k \leq 3$ can ever work.

(a) One period ($k = 1$). The condition is $\delta \geq 3$: impossible for $\delta < 1$. Explicitly, $3 + 3\delta \geq 6 + 2\delta \iff \delta \geq 3$. Cooperation cannot be supported.

(b) Two periods ($k = 2$). The condition is

$$\delta + \delta^2 \geq 3 \iff \delta^2 + \delta - 3 \geq 0.$$

The roots are $\delta = \frac{-1 \pm \sqrt{13}}{2} \approx -2.30,\ 1.30$; the quadratic is negative on $(0,1)$ (at most $1+1-3 = -1$). Cooperation cannot be supported.

(c) Ten periods ($k = 10$). The condition is

$$\delta + \delta^2 + \dots + \delta^{10} \geq 3.$$

The left side rises from 0 to 10 on $[0,1]$, so there is exactly one root in $(0,1)$. Numerically (bisection or a polynomial root-finder),

$$\delta_{10} \approx 0.7627.$$

Cooperation is an SPE outcome iff $\delta \geq \delta_{10}$. For comparison, the thresholds for other lengths are $\delta_4 \approx 0.888$, $\delta_5 \approx 0.834$, $\delta_6 \approx 0.805$, $\delta_8 \approx 0.775$, $\delta_{20} \approx 0.7506$, and grim trigger ($k = \infty$) needs $\frac{\delta}{1-\delta} \geq 3$, i.e. $\delta \geq \frac{3}{4}$ ((10.5): $\frac{6-3}{6-2} = \frac{3}{4}$).

Step 3: the punishment phase. In a punishment period the prescribed action is C, a stage best reply to C. Deviating to NC pays $0 < 2$ today and either leaves the future unchanged or, with the restart rule, lengthens the punishment, which pays $2 < 3$ per period. Either way the deviation loses, for every δ. So the only binding constraint is Step 2's, and the profile is an SPE exactly when that holds.

(d) Interpretation. A punishment deters only if the deviator loses more than he grabs. Here the grab is 3 and each period of punishment costs only 1, so at least four periods are needed, and even then only very patient players are deterred. Longer punishments lower the threshold, with grim trigger as the limit. Short, forgiving punishments are attractive (mistakes are not punished forever), but they need a punishment phase that hurts much more per period than reverting to the Nash equilibrium does here.

(e) Verification. `check_C_L05.py` confirms that (C,C) is the unique stage equilibrium and derives the condition $\sum_{j=1}^{k} \delta^j \geq 3$ symbolically for $k = 1, 2, 3, 4, 10$. It shows (a), (b), and also $k = 3$ fail on the grid $\delta = j/1000$. For (c) it finds the single root in $(0,1)$, $0.762700984\ldots$, and brackets it in exact fractions: the condition fails at 0.762700, holds at 0.762701, fails at 0.76 and holds at 0.77. It also checks that the thresholds decrease in k towards $\frac{3}{4}$. Output: "Munoz-Garcia 6.1: k=1 and k=2 never work (k=3 needs delta=1); k=10: delta >= 0.762701 (book: 0.76, which fails); grim: 3/4; k=4: 0.8882". *Comparison with Muñoz-Garcia's answer* (pp. 275–278): the same conditions and conclusions for (a) and (b) ($\delta \geq 3$; $\delta^2 + \delta - 3 \geq 0$ with roots -2.3 and 1.3). For (c) the book reports $\delta \geq 0.76$, which is the threshold rounded *down*: at $\delta = 0.76$ the sum is about $2.963 < 3$, so cooperation fails. The correct statement is $\delta \geq 0.7627$. Two slips in the book's text for (a): it says the punisher finds "NC a best response to player i playing NC", meaning C against C; and it calls (NC, NC) the "competitive outcome" where it means the cooperative one. The book does not check the punishment phase.

(f) Common mistakes.

- Calling these strategies "tit-for-tat". True tit-for-tat copies the opponent's last move, which is not the same as "both play C for k periods". Tit-for-tat proper is generally *not* subgame perfect, because after a deviation it requires a player to punish while being punished. The book uses the name loosely.
- Comparing the streams only up to period 1 ("$6 + 2\delta$ versus $3 + 3\delta$") when $k > 1$. All k punishment periods enter the comparison.
- Rounding the threshold to 0.76. A threshold must be rounded up, or stated as the exact root.

Answer. With a k-period reversion to (C,C), cooperation is an SPE iff $\sum_{j=1}^{k} \delta^j \geq 3$ (the punishment phase never binds). (a) $k = 1$: $\delta \geq 3$, impossible. (b) $k = 2$: $\delta^2 + \delta - 3 \geq 0$, root $\frac{-1+\sqrt{13}}{2} \approx 1.30 > 1$, impossible. (c) $k = 10$: $\delta + \delta^2 + \cdots + \delta^{10} \geq 3$, i.e. $\delta \geq \delta_{10} \approx 0.7627$ (the book's 0.76 is slightly too low). Grim trigger would need $\delta \geq \frac{3}{4}$.

Item 4 (item 44 of the combined list)**: Muñoz-Garcia Ch. 6 Exercise 9** Exam level · difficulty 3/5

Source: Muñoz-Garcia Ch. 6 Exercise 9, page 305. The statement is reproduced verbatim in the list (item 44); it is not repeated here.

(a) What is asked. For the infinitely repeated 2×2 game of Table 6.6: (a) the stage Nash equilibrium; (b) the feasible payoff set and the normalization used; (c) the feasible, individually rational set; (d) a Nash equilibrium that gives $(9, 9)$ every period, and the condition it needs on δ.

(b) Method. Stage equilibrium by strict dominance (Tadelis §4.1). Feasible payoffs as the convex hull of the stage payoffs (Tadelis Defs. 10.6–10.7, p. 209–210) with payoffs measured as average discounted payoffs (Def. 10.3, p. 194). Individual rationality relative to the minimax payoff, and the folk theorem (Theorem 10.1, p. 211; MWG Prop. 12.AA.5, p. 422, for the minimax version). Part (d) uses grim trigger and the threshold (10.5), p. 202, checked by the one-stage deviation principle (Prop. 10.3, p. 198). Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22. Notation as in `Notas/L05/L05-06-folk-theorem.md`.

(c) Solution. Notation. $q = \Pr(L)$ in a mixed strategy of player 2, $p = \Pr(U)$ for player 1. The average payoff of a stream is $\bar{v}_i = (1-\delta)\sum_{t\geq 1} \delta^{t-1} u_i(a^t)$. Best replies are underlined:

	L	R
U	$9,\ 9$	$1,\ \underline{10}$
D	$\underline{10},\ 1$	$\underline{7},\ \underline{7}$

(a) Stage equilibrium. For player 1, D beats U against both columns ($10 > 9$, $7 > 1$), so D is strictly dominant; by symmetry R is strictly dominant for player 2. The only outcome consistent with Nash equilibrium play is (D, R), payoffs $(7, 7)$. There is no mixed equilibrium: for every q, $u_1(D, q) = 10q + 7(1-q) = 7 + 3q$ exceeds $u_1(U, q) = 9q + 1(1-q) = 1 + 8q$, because $(7 + 3q) - (1 + 8q) = 6 - 5q > 0$ on $[0, 1]$. The game is a Prisoner's Dilemma: (U, L) gives both $9 > 7$.

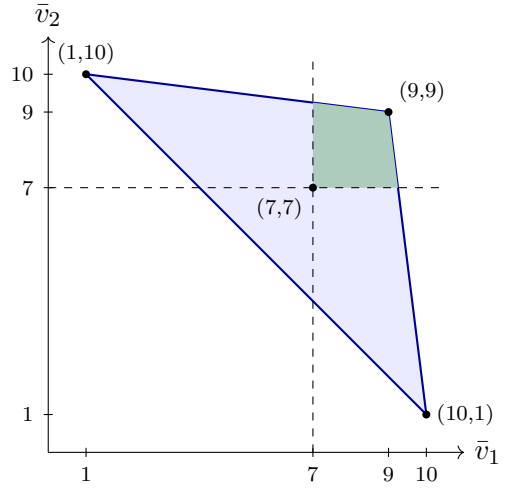
(b) Feasible payoffs. *Normalization.* Repeated-game payoffs are measured as average discounted payoffs, $\bar{v}_i = (1-\delta)v_i$. With this scaling a constant stream x is worth x (Tadelis (10.2), p. 195), so repeated-game payoffs are in the same units as stage payoffs and can be drawn in the same plane. *Convexification.* The four pure payoff vectors are $(9, 9)$, $(1, 10)$, $(10, 1)$, $(7, 7)$. Every point of their convex hull is reached by playing the outcomes with the corresponding long-run frequencies (exactly with a public randomization device, as in the book, or approximately for δ near 1 with deterministic sequences). The feasible set is

$$V = \text{CoHull}\{(9, 9), (1, 10), (10, 1), (7, 7)\}.$$

Its shape. $(7, 7)$ lies strictly inside the triangle with vertices $(1, 10)$, $(10, 1)$, $(9, 9)$, since

$$(7, 7) = \tfrac{3}{7}(9, 9) + \tfrac{2}{7}(1, 10) + \tfrac{2}{7}(10, 1),$$

with all weights positive. So V is that triangle. Its edges are: $v_1 + v_2 = 11$ from $(1, 10)$ to $(10, 1)$; $v_2 = 10 - \frac{v_1 - 1}{8}$ from $(1, 10)$ to $(9, 9)$; and $v_1 = 10 - \frac{v_2 - 1}{8}$ from $(10, 1)$ to $(9, 9)$.



Feasible set V (triangle) and the FIR set (shaded, without its edges on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$).

(c) Individually rational payoffs. Player 1's minimax payoff is

$$\underline{v}_1 = \min_{q\in[0,1]} \max\{u_1(U,q),\, u_1(D,q)\} = \min_{q} (7+3q) = 7 \quad (q=0,\ \text{i.e. } R),$$

because D is the best reply for every q. By symmetry $\underline{v}_2 = 7$ (player 1 plays D). Here the minimax payoff equals the stage Nash payoff. The feasible, individually rational (FIR) set is

$$\{\bar{v} \in V : \bar{v}_1 > 7,\ \bar{v}_2 > 7\}.$$

Cutting the triangle with $\bar{v}_1 = 7$ (on the edge $(1,10)$–$(9,9)$: $\bar{v}_2 = 10 - \frac{6}{8} = \frac{37}{4}$) and with $\bar{v}_2 = 7$ (on the edge $(10,1)$–$(9,9)$: $\bar{v}_1 = \frac{37}{4}$) gives the quadrilateral with corners

$$(7,7), \quad (\tfrac{37}{4},7), \quad (9,9), \quad (7,\tfrac{37}{4}),$$

without its two sides on $\bar{v}_1 = 7$ and $\bar{v}_2 = 7$. Its area is $\frac{9}{2}$.

(d) A Nash equilibrium with $(9,9)$ every period. Grim trigger:

- player 1 plays U in period 1 and as long as every past outcome was (U,L); after any other history, D forever;
- player 2 plays L in period 1 and as long as every past outcome was (U,L); after any other history, R forever.

Cooperation phase. Staying gives $9 + 9\delta + 9\delta^2 + \cdots = \frac{9}{1-\delta}$. The best deviation is D (10 against L), followed by (D,R) forever: $10 + \frac{7\delta}{1-\delta}$. No deviation pays iff

$$\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta} \iff 9 \geq 10(1-\delta) + 7\delta = 10 - 3\delta \iff \delta \geq \frac{1}{3}.$$

(In average payoffs: $9 \geq (1-\delta)10 + \delta\, 7$.) By (10.5), $\delta^* = \frac{10-9}{10-7} = \frac{1}{3}$. Player 2's problem is identical by symmetry. *Punishment phase.* (D,R) is a stage equilibrium and the continuation does not depend on today's action, so deviating to U (or L) loses $7 - 1 = 6$ today and changes nothing later. Both checks pass, so the profile is a subgame-perfect equilibrium, and therefore a Nash equilibrium, for $\delta \geq \frac{1}{3}$.

The restriction is necessary. In any Nash equilibrium with path (U,L) forever, player 1 can deviate to D in period 1 and keep playing D. Since $u_1(D,\cdot) \geq 7$, this gives at least $10 + \frac{7\delta}{1-\delta}$ whatever player 2 does. So $\delta \geq \frac{1}{3}$ is needed for every Nash equilibrium, not just for grim trigger: the punishment already pushes the deviator down to his minimax payoff of 7.

(d) Interpretation. $(9, 9)$ is feasible and gives each player more than the minimax 7, so the folk theorem says it can be sustained by patient enough players. The exercise puts a number on "patient enough": the one-period gain from defecting is 1, the per-period loss from punishment is $9 - 7 = 2$, so $\delta^* = 1/(1+2) = \frac{1}{3}$. The FIR quadrilateral also contains asymmetric outcomes, e.g. near $(\frac{37}{4}, 7)$, which favour player 1 heavily. The folk theorem says these are sustainable too, which is why it explains how cooperation is possible rather than predicting which outcome occurs.

(e) Verification. `check_C_L05.py` confirms that (D, R) is the unique stage equilibrium and that $u_1(D, q) > u_1(U, q)$ for every q on a grid of step $\frac{1}{1000}$, with $\min_q \max = 7$. With sympy it computes the convex hull of the four payoff vectors (the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$, with $(7, 7)$ strictly inside), the corners $(7, \frac{37}{4})$ and $(\frac{37}{4}, 7)$ and the area $\frac{9}{2}$. It solves $\frac{9}{1-\delta} = 10 + \frac{7\delta}{1-\delta}$ to get $\delta = \frac{1}{3}$ and tests the inequality on the grid $k/1000$ and at $\frac{1}{3} \pm \frac{1}{1000}$. Output: "Munoz-Garcia 6.9: stage NE (D,R) (7,7) = minimax; hull = triangle (1,10),(10,1),(9,9); FIR corners (7,7),(37/4,7),(9,9),(7,37/4); delta* = 1/3 (book: 1/3)". *Comparison with Muñoz-Garcia's answer* (pp. 305–309): the same stage equilibrium, minimax $(7, 7)$, FIR set and threshold $\delta \geq \frac{1}{3}$, and the book also shows the strategies are subgame perfect. Three slips in the book: (i) it calls the feasible set "not convex" (p. 307). The convex hull is convex by definition; what is not convex is the set of the four pure payoff vectors, which is what the public randomization device fills in. (ii) It writes $u_2(L; p) = 9p + 1(1-p) = 7 + 3p$ (p. 306); the sum is $1 + 8p$, and $7 + 3p$ is $u_2(R; p)$. (iii) It refers to "Fig. 6.7" for the figure it has just drawn (Fig. 6.11). The book takes the public randomization device as the normalization of part (b); the average-payoff scaling above is the normalization usually meant, and both are needed to draw V in stage-payoff units.

(f) Common mistakes.

- Drawing the four points, or the quadrilateral with $(7, 7)$ as a vertex, as the feasible set. $(7, 7)$ is inside the triangle, so it is not a vertex of V; it is, however, a corner of the FIR set.
- Using ≥ 7 instead of > 7 for individual rationality, or using pure-strategy minimax without checking mixed strategies. Here both give 7, but the definition uses mixed strategies.
- Mixing present values and averages in one inequality, e.g. $9 \geq 10 + \frac{7\delta}{1-\delta}$. Both sides must use the same scaling.

Answer. (a) D and R are strictly dominant: the only stage equilibrium is (D, R), payoffs $(7, 7)$. (b) With average payoffs $(1-\delta)\sum \delta^{t-1} u_i$ and public randomization, V is the convex hull of the four payoff vectors, the triangle $(1, 10)$, $(10, 1)$, $(9, 9)$; $(7, 7)$ is inside it. (c) Minimax $= (7, 7)$; FIR set $= \{\bar{v} \in V : \bar{v}_1 > 7, \bar{v}_2 > 7\}$, the quadrilateral $(7, 7)$, $(\frac{37}{4}, 7)$, $(9, 9)$, $(7, \frac{37}{4})$. (d) Grim trigger (play (U, L) until any deviation, then (D, R) forever) is an SPE, hence a Nash equilibrium, iff $\frac{9}{1-\delta} \geq 10 + \frac{7\delta}{1-\delta}$, i.e. $\delta \geq \frac{1}{3}$ (book: $\frac{1}{3}$). The same bound is necessary for any Nash equilibrium.

Item 5 (item 45 of the combined list)**: Tadelis Exercise 10.3** Exam level · difficulty 3/5

Source: Tadelis Exercise 10.3, page 215–216. The statement is reproduced verbatim in the list (item 45); it is not repeated here.

(a) What is asked. In Tadelis's repeated Prisoner's Dilemma, find the critical discount factor for (M, m) every period under a length-T punishment, (a) for $T = 1$ and (b) for $T = 2$, and (c) compare the two and explain the difference.

(b) Method. Length-T punishments checked with the one-stage deviation principle (Prop. 10.3, p. 198), against the grim-trigger benchmark (10.3), p. 200 ($\delta \geq \frac{1}{4}$ in this game) and (10.5), p. 202. The cooperation-phase test compares the two streams only over the periods in which they differ. Lecture 5 has no slides yet; the repeated-game set-up is Lecture 4, slides 21–22, and the SPE definition is slide 17. Notation as in `Notas/L05/L05-04-grim-trigger-spe.md` and `L05-05-critical-discount-factor.md`.

(c) Solution. Notation. Stage payoffs: $(M, m) = (4, 4)$, $(M, f) = (-1, 5)$, $(F, m) = (5, -1)$, $(F, f) = (1, 1)$. F and f are strictly dominant ($5 > 4$, $1 > -1$), so (F, f) with $(1, 1)$ is the unique stage equilibrium. Strategies with a length-T punishment:

- *cooperation phase*: play M (player 1) and m (player 2);
- *punishment phase*: after any deviation from the prescribed actions, play (F, f) for T periods, then return to the cooperation phase (a deviation during the punishment restarts it).

Present values are measured from the period of the deviation (period 0).

Step 1: the general cooperation-phase condition. Cooperating gives 4 in periods $0, 1, \ldots, T$; deviating gives 5 in period 0 and 1 in periods $1, \ldots, T$. From period $T + 1$ on both streams give 4 forever. So no deviation pays iff

$$4 + 4\sum_{j=1}^{T} \delta^j \;\geq\; 5 + \sum_{j=1}^{T} \delta^j \iff 3\sum_{j=1}^{T} \delta^j \;\geq\; 1.$$

The one-period gain is $5 - 4 = 1$ and each punishment period costs $4 - 1 = 3$.

Step 2: the punishment phase. The prescribed action F is a stage best reply to f. Deviating to M pays $-1 < 1$ today and does not shorten the punishment (with the restart rule it lengthens it, and punishment periods pay $1 < 4$). So no one-stage deviation pays in the punishment phase, for any δ and any T. Only Step 1 binds.

(a) $T = 1$**.** Using present values, as the book's hint suggests:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta \cdot 1 + \frac{4\delta^2}{1-\delta}.$$

Multiplying by $1 - \delta > 0$: $4 \geq 5(1-\delta) + \delta(1-\delta) + 4\delta^2 = 5 - 4\delta + 3\delta^2$, i.e. $3\delta^2 - 4\delta + 1 \leq 0$, i.e. $(3\delta - 1)(\delta - 1) \leq 0$. With $\delta < 1$ this is $\delta \geq \frac{1}{3}$. Step 1 gives the same: $3\delta \geq 1$. So

$$\delta_1 = \frac{1}{3}.$$

(b) $T = 2$**.** Present values:

$$\frac{4}{1-\delta} \;\geq\; 5 + \delta + \delta^2 + \frac{4\delta^3}{1-\delta}.$$

Multiplying by $1-\delta$: $4 \geq 5(1-\delta) + (\delta+\delta^2)(1-\delta) + 4\delta^3 = 5 - 4\delta + 3\delta^3$, i.e.

$$3\delta^3 - 4\delta + 1 \;\leq\; 0.$$

This is the hint's cubic $a\delta^3 - (a+1)\delta + 1$ with $a = 3$. Since $\delta = 1$ is a root,

$$3\delta^3 - 4\delta + 1 = (\delta - 1)(3\delta^2 + 3\delta - 1).$$

For $\delta < 1$ the first factor is negative, so the inequality holds iff $3\delta^2 + 3\delta - 1 \geq 0$ (the same as Step 1: $3(\delta + \delta^2) \geq 1$). The positive root is

$$\delta_2 = \frac{-3 + \sqrt{9+12}}{6} = \frac{\sqrt{21} - 3}{6} \approx 0.2638,$$

and (M, m) in every period is sustained iff $\delta \geq \delta_2$. (The other root, $\frac{-3-\sqrt{21}}{6} < 0$, is irrelevant.)

(c) Comparison.

$$\delta_1 = \tfrac{1}{3} \approx 0.333 \;>\; \delta_2 \approx 0.264 \;>\; \delta_\infty = \tfrac{1}{4},$$

where δ_∞ is the grim-trigger threshold ((10.3): $\frac{5-4}{5-1} = \frac{1}{4}$). A longer punishment costs the deviator more: with $T = 1$ the loss is 3δ, with $T = 2$ it is $3(\delta + \delta^2)$, with grim trigger $3\delta/(1-\delta)$. The gain is 1 in every case. A bigger loss covers the same gain at a lower δ, so the critical value falls as T grows, and it falls by less and less because later punishment periods are discounted more (here $\delta_1 - \delta_2 \approx 0.070$ but $\delta_2 - \delta_\infty \approx 0.014$).

(d) Interpretation. Punishments need not be eternal. In this game the stage gain from cheating is small relative to the loss of a punished period (1 against 3), so even a one-period punishment works for moderately patient players, and two periods already get close to what grim trigger achieves. Short punishments are less costly if a "deviation" was in fact a mistake or a misread signal, since cooperation resumes after T periods. The price is a higher patience requirement.

(e) Verification. `check_C_L05.py` derives the condition $3\sum_{j=1}^{T}\delta^j \geq 1$ symbolically for $T = 1, 2, 5$. It checks $\delta_1 = \frac{1}{3}$ at the threshold and at $\pm\frac{1}{1000}$. It rebuilds the book's cubic from the present values ($3\delta^3 - 4\delta + 1$), checks the factorization and the root $\frac{\sqrt{21}-3}{6} = 0.263763$, brackets that root to 10^{-6} in exact fractions, tests the grid $\delta = j/1000$, and checks the ordering $\frac{1}{3} > \delta_2 > \frac{1}{4}$. Output: "Tadelis 10.3: delta_1 = 1/3; delta_2 = (sqrt(21) - 3)/6 = 0.263763; grim 1/4". Tadelis prints no answer.

(f) Common mistakes.

- Forgetting that cooperation *resumes*: writing the deviation stream as $5 + \delta + \delta^2$ and stopping there, or as $5 + \frac{\delta}{1-\delta}$ (which is grim trigger).
- Keeping the root $\delta = 1$ of the cubic as the answer. It is the trivial root that the hint tells you to factor out; the threshold is the root of the quadratic factor.
- Saying that a shorter punishment is "more lenient, so cooperation is easier". It is the opposite: a weaker threat needs more patience.

Answer. With a length-T reversion to (F, f), (M, m) forever is an SPE iff $3\sum_{j=1}^{T}\delta^j \geq 1$ (the punishment phase never binds). (a) $T = 1$: $\delta_1 = \frac{1}{3}$. (b) $T = 2$: $3\delta^3 - 4\delta + 1 = (\delta - 1)(3\delta^2 + 3\delta - 1) \leq 0$, so $\delta_2 = \frac{\sqrt{21}-3}{6} \approx 0.264$. (c) $\delta_1 > \delta_2 > \frac{1}{4}$ (grim trigger): a longer punishment makes deviating costlier for the same one-period gain, so less patience is needed, with diminishing effect as T grows.

Item 6 (item 46 of the combined list)**: Tadelis Exercise 9.1** Exam level · difficulty 4/5

Source: Tadelis Exercise 9.1, page 186–187. The statement is reproduced verbatim in the list (item 46); it is not repeated here.

(a) What is asked. For the 3×3 game played twice with the first-period outcome observed: (a) all pure-strategy SPE with $\delta = 1$, with history-contingent strategies; (b) for each, the smallest δ that supports it.

(b) Method. Multistage games (Tadelis §9.1–9.4, pp. 175–184) and the finitely repeated game $G(2, \delta)$ (§10.1, pp. 190–192). Period-2 play must be a stage Nash equilibrium after every history (Prop. 9.2, p. 181). A first-period profile that is not a stage equilibrium can be supported only by a carrot (a good stage equilibrium on the path) and a stick (a bad one after a deviation), and only when $\delta \geq \delta^* = g/(c - s)$ (`Notas/L04/L04-02`, `Notas/L05/L05-02`; Tadelis's own two-stage example, p. 191). Deviations are checked one stage at a time (one-stage deviation principle, Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, payoff $\sum_t \delta^{t-1} u_i^t$, each period starts a subgame) and slide 17 (SPE).

(c) Solution. Notation. $v_i = u_i(a^1) + \delta\, u_i(a^2)$, where a^t is the period-t action profile. A pure strategy for player i is a period-1 action and a map from the nine possible period-1 outcomes to a period-2 action. $g_i(a)$ is player i's best one-shot gain from deviating from a:

$$g_1(a) = \max_r u_1(r, a_2) - u_1(a), \qquad g_2(a) = \max_c u_2(a_1, c) - u_2(a).$$

Best replies are underlined:

	L	C	R
T	10, 10	2, 12	0, $\underline{13}$
M	12, 2	$\underline{5}$, $\underline{5}$	0, 0
B	$\underline{13}$, 0	0, 0	$\underline{1}$, $\underline{1}$

Step 1: stage equilibria. Player 1's best replies: to L, B (13); to C, M (5); to R, B (1). Player 2's: to T, R (13); to M, C (5); to B, R (1). The mutual best replies are

$$(M, C) \text{ with } (5, 5) \quad \text{(the carrot)}, \qquad (B, R) \text{ with } (1, 1) \quad \text{(the stick)}.$$

Each player gets 5 in the carrot and 1 in the stick, so the largest punishment one period can deliver is $c - s = 5 - 1 = 4$.

Step 2: period 2 (the last phase). After any first-period outcome the subgame is the stage game (plus a sunk constant), so it must be (M, C) or (B, R). No one-stage deviation in period 2 pays, because both are stage equilibria and nothing follows.

Step 3: period 1 (which profiles can be supported). A first-period profile a is supportable if and only if each player's gain is at most what the threat can take away:

$$g_i(a) \;\leq\; \delta\,(5 - 1) = 4\delta, \qquad i = 1, 2, \tag{1}$$

using (M, C) after a and (B, R) after any deviation. No continuation can do better, since 5 is the most and 1 the least a player can get in period 2. For a stage equilibrium a, $g_i(a) = 0$ and any continuation works. The gains are:

a	$u(a)$	$g_1(a)$	$g_2(a)$	supportable ($\delta = 1$)?	$\delta^* = \max_i g_i/4$
(T, L)	$(10, 10)$	$13 - 10 = 3$	$13 - 10 = 3$	yes	$3/4$
(T, C)	$(2, 12)$	$5 - 2 = 3$	$13 - 12 = 1$	yes	$3/4$
(T, R)	$(0, 13)$	$1 - 0 = 1$	0	yes	$1/4$
(M, L)	$(12, 2)$	$13 - 12 = 1$	$5 - 2 = 3$	yes	$3/4$
(M, C)	$(5, 5)$	0	0	yes	0
(M, R)	$(0, 0)$	$1 - 0 = 1$	$5 - 0 = 5$	no	–
(B, L)	$(13, 0)$	0	$1 - 0 = 1$	yes	$1/4$
(B, C)	$(0, 0)$	$5 - 0 = 5$	$1 - 0 = 1$	no	–
(B, R)	$(1, 1)$	0	0	yes	0

(a) The pure SPE with $\delta = 1$. There are nine SPE *paths*:

- the four sequences of stage equilibria, played regardless of history (Prop. 9.1, p. 180): (M, C) then (M, C); (M, C) then (B, R); (B, R) then (M, C); (B, R) then (B, R);
- the five carrot-and-stick paths: a then (M, C), for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$.

For a carrot-and-stick path the strategies are, for $\hat{a} = (\hat{a}_1, \hat{a}_2)$ one of those five:

$$s_1 : \text{ play } \hat{a}_1; \text{ in period 2 play } M \text{ if } \hat{a} \text{ was played, } B \text{ otherwise;}$$
$$s_2 : \text{ play } \hat{a}_2; \text{ in period 2 play } C \text{ if } \hat{a} \text{ was played, } R \text{ otherwise.}$$

One-stage deviation checks for each path (with $\delta = 1$; path payoff against best deviation):

$\hat{a}$	P1: path	P1: best deviation	P2: path	P2: best deviation
(T, L)	$10 + 5 = 15$	B: $13 + 1 = 14$	$10 + 5 = 15$	R: $13 + 1 = 14$
(T, C)	$2 + 5 = 7$	M: $5 + 1 = 6$	$12 + 5 = 17$	R: $13 + 1 = 14$
(T, R)	$0 + 5 = 5$	B: $1 + 1 = 2$	$13 + 5 = 18$	none (best reply)
(M, L)	$12 + 5 = 17$	B: $13 + 1 = 14$	$2 + 5 = 7$	C: $5 + 1 = 6$
(B, L)	$13 + 5 = 18$	none (best reply)	$0 + 5 = 5$	R: $1 + 1 = 2$

In period 2 both players play a stage equilibrium after every history, so no deviation pays there. (M, R) and (B, C) cannot be supported: a gain of 5 exceeds the largest punishment, 4. For instance, at (M, R) player 2 gets $0 + 5 = 5$ on the path but $5 + 1 = 6$ by deviating to C.

These strategies are not the only ones. Any assignment of (M, C) or (B, R) to the nine period-1 outcomes that satisfies (1) for the deviations that would be profitable also works (e.g. for (T, R) only player 1's deviation to B must be punished). Counting every such map, `check_C_L05.py` finds 992 pure SPE strategy profiles at $\delta = 1$, all with one of the nine paths above.

(b) The smallest discount factor. From (1), $\delta^*(a) = \max_i g_i(a)/4$ (last column of the first table). Solving each binding inequality:

- (T, L): $10 + 5\delta \geq 13 + \delta \iff 4\delta \geq 3 \iff \delta \geq \frac{3}{4}$ (both players);
- (T, C): player 1 binds, $2 + 5\delta \geq 5 + \delta \iff \delta \geq \frac{3}{4}$; player 2 needs $12 + 5\delta \geq 13 + \delta \iff \delta \geq \frac{1}{4}$;
- (M, L): the mirror image of (T, C), $\delta \geq \frac{3}{4}$;
- (T, R): only player 1, $0 + 5\delta \geq 1 + \delta \iff \delta \geq \frac{1}{4}$;
- (B, L): only player 2, $0 + 5\delta \geq 1 + \delta \iff \delta \geq \frac{1}{4}$;
- the four sequences of stage equilibria: any $\delta \geq 0$.

Below δ^* no continuation map works, because (M, C) after $\hat{a}$ and (B, R) after the deviation is already the largest reward gap available.

A caveat: renegotiation. The stick (B, R) is Pareto-dominated by (M, C). After a deviation both players would like to "forgive" and play (M, C) in period 2, and if they can renegotiate the threat fails (Gibbons §2.3.A, pp. 87–88; `Notas/L05/L05-02`). The SPE above take the threat as given.

(d) Interpretation. Two stage equilibria with different payoffs give the second period a carrot and a stick. That lets the players reach (T, L), which pays $(10, 10)$, twice the $(5, 5)$ of the good stage equilibrium, in the first period. Profiles with a gain above 4 cannot be enforced by a single future period, however patient the players are. This is the finite-horizon version of the logic of Lecture 5: what can be sustained today is limited by how much the future can reward and punish. With a unique stage equilibrium (item Cabral 7.13) that gap is zero and nothing can be sustained.

(e) Verification. `check_C_L05.py` finds the two stage equilibria and enumerates all 9×2^9 combinations of first-period profile and continuation map at $\delta = 1$ (992 SPE, 9 paths; (M, R) and (B, C) excluded). For every threshold it checks that each path is an SPE at δ^* and at $\delta^* + \frac{1}{1000}$, that no continuation map at all supports it at $\delta^* - \frac{1}{1000}$, and that on the grid $\delta = k/1000$ the carrot-and-stick map works exactly when $\delta \geq \delta^*$. Output: "Tadelis 9.1: stage NE (M,C),(B,R); 992 pure SPE at delta=1 on 9 paths; delta* = 3/4 (TL, TC, ML), 1/4 (TR, BL); grid and +-1/1000 pass". Tadelis prints no answer.

(f) Common mistakes.

- Listing only the outcome "(T, L) then (M, C)" without the off-path play. The question asks for history-contingent strategies, and the threat to play (B, R) after a deviation is the whole point.
- Using the smallest gain instead of the largest: the threshold is $\max_i g_i/4$, so in (T, C) player 1's gain of 3 binds, not player 2's gain of 1.
- Claiming (M, R) or (B, C) can be sustained with enough patience. With one period left the punishment is capped at $5 - 1 = 4$, which is below the gain of 5 even at $\delta = 1$.

Answer. Stage equilibria: (M, C) $(5, 5)$ and (B, R) $(1, 1)$. (a) Pure SPE paths with $\delta = 1$: the four sequences of stage equilibria, and a then (M, C) for $a \in \{(T, L), (T, C), (T, R), (M, L), (B, L)\}$, supported by "play (M, C) after a, (B, R) after any other outcome". (M, R) and (B, C) cannot be supported. (b) $\delta^* = \frac{3}{4}$ for (T, L), (T, C), (M, L); $\frac{1}{4}$ for (T, R), (B, L); 0 for the sequences of stage equilibria.

Item 7 (item 47 of the combined list)**: Gibbons Problem 2.11** Exam level · difficulty 4/5

Source: Gibbons Problem 2.11, page 135. The statement is reproduced verbatim in the list (item 47); it is not repeated here.

(a) What is asked. For the 3×3 game played twice without discounting, decide whether $(4, 4)$ can be obtained in the first stage of a pure-strategy SPE, and give the strategies or a proof.

(b) Method. Two-stage repeated game with several stage equilibria (Gibbons §2.3.A, pp. 82–88, especially the three-strategy example of pp. 84–87; Tadelis §9.4, pp. 180–184, and §10.1, p. 191). In the second stage a stage Nash equilibrium must be played after every history (Tadelis Prop. 9.2, p. 181). A first-stage profile that is not a stage equilibrium is supported by a *reward* equilibrium after it and a *punishment* equilibrium after deviations. Deviations are checked one stage at a time (Tadelis Thm. 9.1, p. 185). Slides: Lecture 4, slide 22 ($G(T, \delta)$, each period starts a subgame) and slide 17 (SPE). Notation as in `Notas/L05/L05-02-two-stage-carrot-stick.md`.

(c) Solution. Notation. $v_i = u_i(a^1) + u_i(a^2)$ (no discounting). Best replies are underlined:

	L	C	R
T	$\underline{3},\ \underline{1}$	$0,\ 0$	$\underline{5},\ 0$
M	$2,\ 1$	$\underline{1},\ \underline{2}$	$3,\ 1$
B	$1,\ 2$	$0,\ 1$	$4,\ \underline{4}$

Step 1: stage equilibria. Player 1's best replies: to L, T (3); to C, M (1); to R, T (5). Player 2's: to T, L (1); to M, C (2); to B, R (4). The mutual best replies are

$$(T, L) \text{ with } (3, 1), \qquad (M, C) \text{ with } (1, 2).$$

(B, R) is *not* a stage equilibrium: against R, player 1 prefers T ($5 > 4$). Since the question asks for a pure-strategy SPE, second-stage play must be one of these two pure equilibria after every history.

Step 2: who must be deterred. At (B, R), player 2 is already best-responding (4 is the most she can get against B), so only player 1 has a profitable one-stage deviation: to T, gaining $5 - 4 = 1$ (to M he would get $3 < 4$). Player 1 gets 3 in (T, L) and 1 in (M, C), so the second stage can reward him with 3 and punish him with 1: a swing of $2 > 1$.

Step 3: the strategies.

- Player 1: play B in stage 1; in stage 2 play T if the stage-1 outcome was (B, R), and M otherwise.
- Player 2: play R in stage 1; in stage 2 play L if the stage-1 outcome was (B, R), and C otherwise.

Step 4: one-stage deviation checks. *Stage 2 (every history).* The prescribed profile is (T, L) or (M, C), a stage equilibrium, and nothing follows, so no deviation pays.

Stage 1, player 1. On the path he gets $4 + 3 = 7$. Deviating to T gives 5 now and then (M, C), so $5 + 1 = 6 < 7$; deviating to M gives $3 + 1 = 4 < 7$.

Stage 1, player 2. On the path she gets $4 + 1 = 5$. Deviating to L gives 2 now and then (M, C), so $2 + 2 = 4 < 5$; deviating to C gives $1 + 2 = 3 < 5$. (The "punishment" (M, C) is actually better for player 2 than (T, L), but she has nothing to gain in stage 1, so it does not matter.)

All one-stage deviations lose, so the profile is a pure-strategy SPE and the answer is **yes**: $(4, 4)$ is reached in the first stage, with total payoffs $(7, 5)$.

How much room there is. With a discount factor δ, player 1's constraint is $4 + 3\delta \geq 5 + \delta \iff \delta \geq \frac{1}{2}$, so the construction survives some impatience; player 2's constraint, $4+\delta \geq 2+2\delta \iff \delta \leq 2$, holds for every $\delta \in [0, 1]$. Among all $2^9 = 512$ assignments of (T, L) or (M, C) to the nine stage-1 outcomes, exactly $128 = 2^7$ support (B, R) first: those with (T, L) after (B, R) and (M, C) after player 1's deviation to T. The other seven outcomes can be assigned either way.

(d) Interpretation. The second stage offers two equilibria that player 1 ranks differently, and that difference is a credible incentive. The players can coordinate on the efficient (B, R) today because both expect player 1 to be "paid" with his preferred equilibrium (T, L) tomorrow if he behaves, and pushed to (M, C) if he grabs the extra 1. Gibbons (pp. 87–88) warns that such threats may be renegotiated when the punishment equilibrium is Pareto-dominated by the reward equilibrium. That is not the case here: player 2 prefers (M, C) to (T, L), so after player 1's deviation she would not agree to forgive him, and the threat survives that critique.

(e) Verification. `check_C_L05.py` finds the two pure stage equilibria, confirms that player 1's best reply to R is T, and enumerates all 512 continuation maps with (B, R) first at $\delta = 1$. It finds 128 SPE, all with (T, L) after (B, R) and (M, C) after (T, R). It checks the payoffs 7 against 6, and on the grid $\delta = k/1000$ that the constructed profile works exactly for $\delta \geq \frac{1}{2}$. Output: "Gibbons 2.11: stage NE (T,L),(M,C); (B,R) first is a pure SPE: 128 of 512 maps work, all with (T,L) after (B,R) and (M,C) after (T,R)". Gibbons prints no answer. *Comparison with Muñoz-Garcia Ch. 6 Ex. 7* (pp. 299–303, the same game): the same answer (yes, (B, R) then (T, L), with $7 > 6$ for player 1). Two differences. (i) The book punishes with the stage game's mixed equilibrium (each player mixes $\frac{1}{2}$–$\frac{1}{2}$ over $\{T, M\}$ and $\{L, C\}$, payoffs $(\frac{3}{2}, 1)$), giving $5 + \frac{3}{2} = 6.5 < 7$. That works as an SPE, but it is not a *pure-strategy* SPE as the question asks; the pure equilibrium (M, C) is both admissible and a harsher punishment. (ii) The book lists $BR_1(R) = B$; in fact $BR_1(R) = T$ $(5 > 4)$, which is exactly why (B, R) needs support. Its list of pure equilibria, (T, L) and (M, C), is right.

(f) Common mistakes.

- Answering "no, because (B, R) is not a Nash equilibrium of the stage game". In a repeated game with several stage equilibria, non-equilibrium play *can* be supported in early stages.
- Using a mixed equilibrium as the punishment when the question restricts to pure strategies, or a non-equilibrium profile (e.g. (B, C)) in the last stage.
- Checking only player 2 (whose payoff 4 is high) and missing that player 1 is the one tempted.

Answer. Yes. Stage equilibria: (T, L) $(3, 1)$ and (M, C) $(1, 2)$. Strategies: play (B, R) in stage 1; in stage 2 play (T, L) if stage 1 was (B, R) and (M, C) otherwise. Player 1: $4 + 3 = 7 > 5 + 1 = 6$ (deviation to T) and $> 3 + 1$ (to M); player 2: $4 + 1 = 5 > 2 + 2$ (to L) and $> 1 + 2$ (to C); stage 2 is always a stage equilibrium. This is a pure-strategy SPE with $(4, 4)$ in the first stage (it would survive discounting down to $\delta = \frac{1}{2}$).

Item 8 (item 48 of the combined list)**: Tadelis Exercise 11.5** Exam level · difficulty 4/5

Source: Tadelis Exercise 11.5, page 236–237. The statement is reproduced verbatim in the list (item 48); it is not repeated here.

(a) What is asked. In three-period alternating-offer bargaining over a pie of 1, with player 1 proposing first and discount factors $\delta_1 \neq \delta_2$: (a) the SPE outcome; (b) player 1's advantage when $\delta_1 = \delta_2$; (c) when player 2 has the advantage, and why.

(b) Method. Backward induction in finite alternating-offer bargaining (Tadelis §11.2, pp. 224–227; Gibbons §2.1.D, pp. 68–70). In the last period the proposer takes everything. In each earlier period the proposer offers the responder exactly the discounted value of what the responder would get next period, and the responder accepts at indifference (the argument of Prop. 11.2, p. 223; agreement in round one, Prop. 11.3, p. 226). Slides: Lecture 4, slides 9–10 (backward induction) and slide 17 (SPE). Lecture 5's bargaining part has no slides yet. Notation as in `Notas/L04/L04-06-finite-alternating-offers.md`.

(c) Solution. Notation. x_t = player 1's share of the pie when agreement is reached in period $t \in \{1, 2, 3\}$, so player 2 gets $1 - x_t$. Player i's payoff from agreement $(x, 1-x)$ in period t is δ_i^{t-1} times his share. Players 1, 2, 1 propose in periods 1, 2, 3, and no agreement in period 3 gives $(0, 0)$. "Advantage" means a share above $\frac{1}{2}$.

Step 1: period 3 (player 1 proposes, last period). Player 2 compares accepting $1 - x$ with rejecting, which gives 0. She accepts every $x < 1$ strictly. At $x = 1$ she is indifferent, and in an SPE she must accept: if she rejected $x = 1$, player 1 would want the largest $x < 1$, which does not exist, so he would have no best offer. Player 1 offers $x_3 = 1$. Period-3 values: player 1 gets 1, player 2 gets 0; in period-1 terms, $(\delta_1^2,\ 0)$.

Step 2: period 2 (player 2 proposes). Player 1 accepts x iff x is at least what rejecting gives him, namely 1 one period later, worth $\delta_1 \cdot 1$ today:

$$x \ \geq \ \delta_1 \, x_3 = \delta_1. \tag{2}$$

Player 2 keeps the most she can by offering exactly $x_2 = \delta_1$ (by the same argument as in Step 1, player 1 accepts at equality). She then gets $1 - \delta_1$. Making an offer player 1 rejects gives her $\delta_2 \cdot 0 = 0 < 1 - \delta_1$, so she makes the acceptable offer.

Step 3: period 1 (player 1 proposes). Player 2 accepts x iff her share now is at least the value of rejecting, i.e. $1 - x_2$ one period later, discounted by her own δ_2:

$$1 - x \ \geq \ \delta_2 \, (1 - x_2) = \delta_2 \, (1 - \delta_1).$$

Player 1 keeps as much as possible:

$$x_1 = 1 - \delta_2(1 - \delta_1) = 1 - \delta_2 + \delta_1\delta_2. \tag{3}$$

He prefers this to a rejected offer, which gives him $\delta_1 x_2 = \delta_1^2$: indeed $x_1 - \delta_1^2 = (1-\delta_1)(1+\delta_1-\delta_2) > 0$.

(a) The SPE. Strategies at every node:

- player 1: offer $x_1 = 1 - \delta_2(1 - \delta_1)$ in period 1; in period 2 accept iff offered $x \geq \delta_1$; in period 3 offer $x_3 = 1$;
- player 2: in period 1 accept iff $1 - x \geq \delta_2(1 - \delta_1)$; in period 2 offer $x_2 = \delta_1$; in period 3 accept every offer.

Each acceptance rule holds after every possible offer, including offers never made on the path, so every subgame is played optimally. The outcome is immediate agreement:

$$(v_1, v_2) = \big(1 - \delta_2(1-\delta_1),\ \ \delta_2(1-\delta_1)\big).$$

The tie-breaking argument of Step 1 applies at every round, so this SPE is unique.

(b) Equal discount factors. Setting $\delta_1 = \delta_2 = \delta$ in (3):

$$v_1 = 1 - \delta + \delta^2, \qquad v_2 = \delta - \delta^2,$$

$$v_1 - v_2 = 1 - 2\delta + 2\delta^2 = (1-\delta)^2 + \delta^2 \ > \ 0.$$

Player 1 gets strictly more than half for every δ. He proposes both first and last. The edge $(1-\delta)^2 + \delta^2$ is smallest at $\delta = \frac{1}{2}$ (derivative $-2(1-\delta) + 2\delta = 0$), where it equals $\frac{1}{2}$ and $v_1 = \frac{3}{4}$.

(c) When player 2 is ahead. Player 2 gets more than half iff

$$\delta_2\,(1-\delta_1) > \tfrac{1}{2} \iff \delta_1 < 1 - \frac{1}{2\delta_2}.$$

This needs $\delta_2 > \frac{1}{2}$ (otherwise $\delta_2(1-\delta_1) \le \frac{1}{2}$ for every δ_1) and δ_1 low enough. Example: $\delta_2 = 0.9$ gives the cut-off $\delta_1 < 1 - \frac{1}{1.8} = \frac{4}{9}$; with $\delta_1 = 0.2$, $v_2 = 0.9 \times 0.8 = 0.72$. *Why.* Player 1's power comes from the last period. An impatient player 1 values it at only δ_1 from period 2's viewpoint, so player 2 keeps $1 - \delta_1$ when she proposes. A patient player 2 values that period-2 position almost fully (δ_2), so in period 1 player 1 must concede $\delta_2(1-\delta_1)$. Patience is bargaining power:

$$\frac{\partial v_1}{\partial \delta_1} = \delta_2 > 0, \qquad \frac{\partial v_1}{\partial \delta_2} = -(1-\delta_1) < 0.$$

The infinite-horizon formula $v_1 = \frac{1-\delta_2}{1-\delta_1\delta_2}$ (Tadelis §11.3; `Notas/L04/L04-07`) has the same signs.

(d) Interpretation. First-mover and last-mover advantages both favour player 1, so only a large patience gap in player 2's favour can overturn them. As $\delta \to 0$ (equal patience) the first-mover advantage takes over and $v_1 \to 1$. As $\delta \to 1$ the last-mover advantage takes over and again $v_1 \to 1$, because waiting for period 3 costs player 1 nothing. Patience matters because each player's threat to reject is worth what the next round is worth *to him*: the more patient player loses less by waiting, so the other side must pay him more to settle now.

(e) Verification. `check_C_L05.py` redoes the recursion symbolically ($x_1 = 1 - \delta_2 + \delta_1\delta_2$; $v_1 - v_2 = (1-\delta)^2 + \delta^2$ with minimum $\frac{1}{2}$ at $\delta = \frac{1}{2}$; limits 1 at both ends; the two derivatives). It tests the condition in (c) at the boundary $\delta_1 = \frac{4}{9}$ for $\delta_2 = \frac{9}{10}$ and at $\frac{4}{9} \pm \frac{1}{1000}$, confirms that player 2 can never be ahead when $\delta_2 \le \frac{1}{2}$ (grid of step $\frac{1}{100}$), and repeats the backward induction numerically with offers on a grid of step $\frac{1}{1000}$ for three pairs; e.g. $(\delta_1, \delta_2) = (0.9, 0.7)$ gives $x_1 = 0.93$. Output: "Tadelis 11.5: v1 = 1 - d2(1 - d1), v2 = d2(1 - d1); equal delta edge (1-d)^2 + d^2 >= 1/2; P2 ahead iff d2(1-d1) > 1/2 (boundary 4/9 at d2 = 9/10 checked +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Discounting player 2's continuation with δ_1 (or with a common δ). Each player values the future with *his own* factor: player 2's period-1 threat is $\delta_2(1-\delta_1)$.
- Letting player 2 reject at indifference in period 3 (or player 1 in period 2). Then the proposer has no best offer and there is no SPE; acceptance at indifference is forced.

- Answering (b) with "player 1 moves first" only. With three periods he also moves last, which is why his edge does not vanish as $\delta \to 1$.

Answer. (a) Immediate agreement: player 1 gets $1-\delta_2(1-\delta_1)$, player 2 gets $\delta_2(1-\delta_1)$ (offers: $x_3 = 1$, $x_2 = \delta_1$, $x_1 = 1 - \delta_2 + \delta_1\delta_2$). (b) With $\delta_1 = \delta_2 = \delta$: $v_1 - v_2 = (1-\delta)^2 + \delta^2 > 0$ (at least $\frac{1}{2}$). (c) Player 2 is ahead iff $\delta_2(1-\delta_1) > \frac{1}{2}$, which needs player 2 patient ($\delta_2 > \frac{1}{2}$) and player 1 impatient ($\delta_1 < 1 - \frac{1}{2\delta_2}$). Player 1's last-period power is worth little to an impatient player 1, while player 2's period-2 power is worth a lot to a patient player 2.

Item 9 (item 49 of the combined list)**: Tadelis Exercise 10.7** Advanced · difficulty 4/5

Source: Tadelis Exercise 10.7, page 217–218. The statement is reproduced verbatim in the list (item 49); it is not repeated here.

(a) What is asked. In the bartender–customer game: (a) all stage Nash equilibria; (b) all SPE when it is played on 10 evenings without discounting; (c) with infinite repetition, the prices p at which an SPE has $x = 1$ and a purchase every day; (d) the values of δ for which that price range is non-empty.

(b) Method. (a) Strict dominance (Tadelis §4.1). (b) Finite repetition of a game with a unique stage equilibrium: Prop. 10.1, p. 192 (§10.1). (c)–(d) Grim trigger and the one-stage deviation principle (Prop. 10.3, p. 198; (10.3) p. 200; (10.5) p. 202), applied to each player separately because the game is asymmetric, plus the minimax argument showing the punishment is the harshest available (`Notas/L05/L05-05-critical-discount-factor.md`, §1 and §5). Slides: Lecture 4, slides 21–22 (repeated games, $G(T, \delta)$) and slide 17 (SPE). Lecture 5 has no slides yet.

(c) Solution. Notation. The bartender (B) chooses $x \in [0, 1]$; the customer (K) buys with probability $\beta \in [0, 1]$, without seeing x, so within a day the moves are effectively simultaneous. Stage payoffs:

$$u_B(x, \beta) = \beta p - cx, \qquad u_K(x, \beta) = \beta(vx - p),$$

with $v > c > 0$. The price p is fixed by the market; take $p > 0$ (the case $p = 0$ is noted where it matters). The customer learns x only if he buys. $\delta \in (0, 1)$ is the discount factor (the book's "discount rate").

(a) Stage Nash equilibria. For the bartender, $\partial u_B/\partial x = -c < 0$ whatever β is, so $x = 0$ is strictly dominant: a diluted drink is cheaper, and the customer cannot see it before paying. Against $x = 0$, buying gives the customer $v \cdot 0 - p = -p < 0$ and not buying gives 0, so he does not buy ($\beta = 0$). The unique Nash equilibrium is

$$(x, \beta) = (0, 0), \qquad \text{payoffs } (u_B, u_K) = (0, 0).$$

(If $p = 0$ the customer is indifferent, and every $\beta \in [0, 1]$ with $x = 0$ is an equilibrium, all with payoffs $(0, 0)$.) The efficient outcome, $x = 1$ with a purchase, creates surplus $v - c > 0$ but is not an equilibrium: this is a one-sided Prisoner's Dilemma (a trust game).

(b) Ten evenings. The stage game has a unique Nash equilibrium, so by backward induction (Prop. 10.1) the unique SPE is $x = 0$ and no purchase on every evening after every history. Step by step: on evening 10 nothing follows, $x = 0$ is strictly dominant and so the customer does not buy, whatever he knows about earlier evenings. On evening $k < 10$, by induction play from $k + 1$ on is (0, no purchase) whatever happens on evening k, so evening k is the stage game plus a constant, and the same argument applies. The customer's imperfect information about x on days he does not buy does not matter, because $x = 0$ is strictly dominant at every information set. (With $p = 0$ the bartender's payoff on every evening is $-cx$ whatever the customer does, so he still chooses $x = 0$ every evening, and the customer may buy or not.) Payoffs: $(0, 0)$.

(c) Infinite repetition: the price range. Strategies (grim trigger):

- bartender: $x = 1$ on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise $x = 0$ forever;
- customer: buy on day 1 and as long as every past day had $x = 1$ and a purchase; otherwise never buy again.

On the path the customer buys every day, so he observes every x and any dilution is detected; a refusal to buy is seen by the bartender. In the punishment phase the customer no longer observes x, but nothing there depends on it.

Bartender, cooperation phase. Staying gives $p - c$ every day: $\frac{p-c}{1-\delta}$. The customer buys today without seeing x, so the best deviation is $x = 0$ (any $x \in (0, 1)$ earns $p - cx < p$). It pays p today and 0 forever after:

$$\frac{p-c}{1-\delta} \;\geq\; p \iff p - c \geq (1-\delta)p \iff \delta p \geq c \iff p \;\geq\; \frac{c}{\delta}.$$

Customer, cooperation phase. Staying gives $v - p$ every day: $\frac{v-p}{1-\delta}$. The deviation, not buying, gives 0 today and 0 forever:

$$\frac{v-p}{1-\delta} \;\geq\; 0 \iff p \;\leq\; v.$$

Punishment phase. The prescribed play is the stage equilibrium (0, no purchase) and the future does not depend on today's action, so a deviation only changes today's payoff: the bartender would pay $cx > 0$ for nothing, and the customer would pay p for a drink with $x = 0$. Neither gains.

So the grim-trigger profile is an SPE with $x = 1$ and a purchase every day iff

$$\frac{c}{\delta} \;\leq\; p \;\leq\; v.$$

No other SPE does better. The bartender can guarantee himself 0 every day by choosing $x = 0$ (his minimax payoff, reached when the customer stops buying), and the customer can guarantee 0 by not buying. Grim trigger already pushes each deviator to 0, so no other punishment can widen the range: $[c/\delta, v]$ is the set of prices for which *any* SPE with this path exists.

(d) When the range is non-empty.

$$\frac{c}{\delta} \leq v \iff \delta \geq \frac{c}{v}.$$

Because $v > c$, $\frac{c}{v} < 1$, so the condition holds for all $\delta \in [\frac{c}{v}, 1)$. At $\delta = \frac{c}{v}$ the only price is $p = v$, which leaves the customer no surplus. For example, with $c = 1$, $v = 3$, $\delta = \frac{1}{2}$: $p \in [2, 3]$.

(d) Interpretation. Honest service is sustained by the value of the relationship, and that value has to come from the price. The bartender must earn a premium, $p - c \geq \frac{1-\delta}{\delta} c$ above cost, large enough that losing a regular customer outweighs the one-time saving c from watering one drink. A tourist, on a known finite visit, is cheated on every drink and so never buys; a local, valued as a long-term customer, is served full drinks at a price above cost. This is the logic of quality-assuring prices and relational contracts (Cabral §7.3, p. 174): with p too low there is no rent to lose, and with p above v the customer would rather not buy. The more patient the bartender, the smaller the premium needed.

(e) Verification. `check_C_L05.py` checks with sympy that u_B falls in x for every β (so $x = 0$ is strictly dominant) and that the bartender's condition reduces to $\delta p - c \geq 0$. For $(v, c) = (3, 1)$ and $(10, 9)$ and $\delta \in \{\frac{1}{2}, \frac{9}{10}, \frac{c}{v}, \frac{c}{v} + \frac{1}{1000}\}$ it confirms in exact fractions that both incentive constraints hold at $p = c/\delta$ and $p = v$ and fail at $c/\delta - \frac{1}{1000}$ and $v + \frac{1}{1000}$, or that no price works when $c/\delta > v$. On the grid $\delta = k/1000$ it checks that the range is non-empty exactly when $\delta \geq c/v$. Output: "Tadelis 10.7: stage NE x = 0, no purchase (p > 0); SPE price range c/delta <= p <= v; non-empty iff delta >= c/v (checked at c/v and +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Letting the customer react to x on the same day. He pays before he tastes, so the bartender's deviation earns the full price p that day; the payoff from deviating is p, not 0.
- Giving only the bartender's constraint ($p \geq c/\delta$) and forgetting the customer's ($p \leq v$), or the reverse; the game is asymmetric and each player has his own condition.
- In (b), arguing that ten evenings is "long enough" for reputation to work. With a known last evening and a unique stage equilibrium the game unravels completely.

Answer. (a) $x = 0$ is strictly dominant, so for $p > 0$ the unique NE is $x = 0$ and no purchase, payoffs $(0, 0)$ (for $p = 0$, $x = 0$ with any purchase probability). (b) Unique SPE: $x = 0$ and no purchase on all 10 evenings after every history. (c) With grim trigger, the bartender needs $\frac{p-c}{1-\delta} \geq p$, i.e. $p \geq c/\delta$, and the customer needs $p \leq v$; since these punishments are each player's minimax, the range is exactly $\frac{c}{\delta} \leq p \leq v$. (d) It is non-empty iff $\delta \geq c/v$ (which is below 1 because $v > c$).

Item 10 (item 50 of the combined list)**: Gibbons Problem 2.20** Advanced · difficulty 5/5

Source: Gibbons Problem 2.20, page 137–138. The statement is reproduced verbatim in the list (item 50); it is not repeated here.

(a) What is asked. Show that the stationary strategies given in the problem (player 1 offers $(s^*, 1-s^*)$ and accepts iff $s \geq \delta s^*$; player 2 offers $(1-s^*, s^*)$ and accepts iff $1-s \geq \delta s^*$, with $s^* = 1/(1+\delta)$) are first a Nash equilibrium and then a subgame-perfect equilibrium of the infinite-horizon Rubinstein game.

(b) Method. Nash equilibrium: bound each player's payoff over every possible agreement date against the opponent's fixed strategy. Subgame perfection: the one-stage deviation principle (Tadelis Thm. 9.1, p. 185; its infinite-horizon form, Prop. 10.3, p. 198). It applies because payoffs are discounted and bounded, so a deviation's effect far in the future is negligible. Because the strategies ignore history, there are only four kinds of subgame to check. Model: Gibbons §2.1.D, pp. 68–71 (the infinite-horizon value $1/(1+\delta)$ and the Shaked–Sutton argument, pp. 70–71); Tadelis §11.3, pp. 228–229 (eqs. (11.4)–(11.5)). Slides: Lecture 4, slide 17 (SPE: Nash equilibrium in *every* subgame) and slides 9–10 (backward induction). Notation as in `Notas/L04/L04-07-rubinstein-infinite-horizon.md`.

(c) Solution. Notation and two identities. s is always player 1's share. In period t an agreement $(s, 1-s)$ is worth $\delta^{t-1}s$ to player 1 and $\delta^{t-1}(1-s)$ to player 2; perpetual disagreement gives $(0,0)$. Player 1 proposes in odd periods, player 2 in even ones. With $s^* = \frac{1}{1+\delta}$,

$$1 - s^* = \frac{\delta}{1+\delta} = \delta s^*, \qquad 1 - \delta s^* = \frac{1}{1+\delta} = s^*. \tag{4}$$

So player 2's offer $(1-s^*, s^*)$ gives player 1 exactly δs^*, his acceptance threshold, and player 1's offer gives player 2 exactly $1-s^* = \delta s^*$, her threshold. On the path, player 1 offers $(s^*, 1-s^*)$ in period 1 and player 2 accepts: payoffs $(s^*, 1-s^*) = \left(\frac{1}{1+\delta}, \frac{\delta}{1+\delta}\right)$.

Part 1: Nash equilibrium. *Player 1's best response.* Fix player 2's strategy. Whatever player 1 does, any agreement happens either in an odd period t, on an offer of player 1 that player 2 accepts, so $1-s \geq \delta s^*$, i.e. $s \leq 1-\delta s^* = s^*$ by (4); or in an even period t, on player 2's only offer, $s = 1-s^* = \delta s^*$. Player 1's payoff is therefore at most

$$\max\Big\{ \max_{t \text{ odd}} \delta^{t-1} s^*, \quad \max_{t \text{ even}} \delta^{t-1}\, \delta s^*, \quad 0 \Big\} \;=\; \max\{s^*,\ \delta^2 s^*,\ 0\} = s^*,$$

reached at $t=1$, and his strategy gets exactly s^*. It is a best response.

Player 2's best response. Fix player 1's strategy. Agreement in an odd period t is on player 1's only offer, giving player 2 $\delta^{t-1}(1-s^*)$. Agreement in an even period t needs an offer player 1 accepts, $s \geq \delta s^*$, so player 2 gets at most $\delta^{t-1}(1-\delta s^*) = \delta^{t-1}s^*$. The best she can do is

$$\max\{\, 1-s^* \ (t=1),\ \ \delta s^* \ (t=2),\ \text{smaller values later}\,\} = 1-s^* = \delta s^*,$$

the two candidates being equal by (4). Her strategy (accept in period 1) gets $1-s^*$, so it is a best response. She is indifferent between accepting now and rejecting to propose next period, which is the property that pins down s^*. So the strategies form a Nash equilibrium.

Part 2: subgame perfection. The strategies do not depend on history, so a subgame is characterized only by whose move it is and, for a responder, by the offer on the table. Up to the common factor δ^{t-1} there are four types:

(i) *player 1 about to propose* (odd period);

(ii) *player 2 responding to an arbitrary offer* $(s, 1-s)$ (odd period);

(iii) *player 2 about to propose* (even period);

(iv) *player 1 responding to an arbitrary offer* $(s, 1-s)$ (even period).

Continuation values. Starting a type-(i) subgame, the strategies give agreement at once on $(s^*, 1-s^*)$. Starting a type-(iii) subgame, they give agreement at once on $(1-s^*, s^*)$: player 2 gets s^* and player 1 gets $1-s^* = \delta s^*$. By the one-stage deviation principle it is enough to check single deviations at each type.

(ii) Player 2 responding to $(s, 1-s)$. Accepting gives $1-s$. Rejecting leads to a type-(iii) subgame next period, where she gets s^*, worth δs^* now. Accepting is optimal iff $1-s \geq \delta s^*$, which is exactly her rule (at equality she is indifferent and accepting is a best reply). No one-stage deviation pays, for *every* offer s, on or off the path.

(iv) Player 1 responding to $(s, 1-s)$. Accepting gives s. Rejecting leads to a type-(i) subgame next period, worth δs^*. Accepting is optimal iff $s \geq \delta s^*$: his rule.

(i) Player 1 proposing. If he offers $s \leq s^*$, player 2 accepts $(1-s \geq 1-s^* = \delta s^*)$ and he gets $s \leq s^*$. If he offers $s > s^*$, player 2 rejects; next period player 2 offers $(\delta s^*, \cdot)$, he accepts, and he gets $\delta \cdot \delta s^* = \delta^2 s^* < s^*$. The best offer is $s = s^*$, as prescribed.

(iii) Player 2 proposing. Symmetrically, offering player 1 $s \geq \delta s^*$ is accepted and leaves her $1-s \leq 1-\delta s^* = s^*$; offering less is rejected; player 1 then proposes $(s^*, 1-s^*)$, she accepts, and she gets $\delta(1-s^*) = \delta^2 s^* < s^*$. Her best offer is $s = 1-s^* = \delta s^*$, as prescribed.

No player gains from a single deviation in any subgame, so by the one-stage deviation principle the strategy profile is a Nash equilibrium in every subgame: it is subgame perfect.

Why s^* and not another number. The two indifference conditions are $1-s_1 = \delta(1-s_2)$ (player 2 indifferent in odd periods) and $s_2 = \delta s_1$ (player 1 indifferent in even periods). Substituting, $s_1 = 1-\delta(1-\delta s_1)$, whose only solution is $s_1 = \frac{1}{1+\delta} = s^*$. The same fixed point appears in Gibbons's backward-induction argument, pp. 70–71. The problem does not ask for uniqueness; Rubinstein (1982), via the Shaked–Sutton argument in Gibbons, shows this is the *only* SPE.

(d) Interpretation. With an infinite horizon there is no last period to anchor backward induction, but the game looks the same every time a given player proposes. Each proposer offers the responder exactly what the responder could get by waiting one period and proposing himself, δs^*, and keeps the rest. The first proposer's advantage is $s^* - (1-s^*) = \frac{1-\delta}{1+\delta}$, the cost of one period of delay, and it vanishes as $\delta \to 1$, where the split tends to $(\frac{1}{2}, \frac{1}{2})$. Off the path the strategies tell each responder to accept anything at least as good as the value of waiting: these threats are credible, which is exactly what subgame perfection requires beyond Nash.

(e) Verification. `check_C_L05.py` verifies the identities (4) and the unique solution $s^* = 1/(1+\delta)$ of $s = 1-\delta(1-\delta s)$ with sympy. Then, for $\delta \in \{\frac{1}{2}, \frac{3}{4}, \frac{9}{10}, \frac{99}{100}\}$ in exact fractions, it checks: each responder accepts at the threshold and switches exactly at threshold $\pm\frac{1}{1000}$; the proposers' best one-stage offers, on a grid of step $\frac{1}{1000}$ plus $s^* \pm \frac{1}{1000}$, are the prescribed ones, and a rejected offer yields $\delta^2 s^*$; and the best agreement date against the opponent's strategy is $t = 1$ for player 1 and $t = 1$ (tied with $t = 2$) for player 2, over 59 periods. Output: "Gibbons 2.20: s* = 1/(1+delta), 1 - s* = delta s*; one-stage deviations by proposers and responders unprofitable for delta in {1/2, 3/4, 9/10, 99/100}; cutoffs +-1/1000 pass". Gibbons prints no answer.

(f) Common mistakes.

- Proving only the Nash part, or checking only on-path behaviour. Subgame perfection needs the responder's rule to be optimal after *every* offer, including offers never made.
- Valuing a rejection by the responder's share of the *current* offer, or forgetting to discount: rejecting today is worth δs^* (the proposer's share next period, one period later), not s^*.
- Applying the one-stage deviation principle without saying why it holds in an infinite game: it needs discounting with bounded payoffs (continuity at infinity).

Answer. With $s^* = \frac{1}{1+\delta}$ we have $1 - s^* = \delta s^*$ and $1 - \delta s^* = s^*$. Nash: against player 2's strategy, player 1 can get at most s^* (odd periods) or $\delta^2 s^*$ (even periods), and his strategy gets s^*; against player 1's strategy, player 2 can get at most $1 - s^* = \delta s^*$, and accepting in period 1 gets it. SPE: in each of the four subgame types (proposer 1, responder 2, proposer 2, responder 1), no one-stage deviation pays, because each responder accepts exactly when the offer is worth at least waiting (δs^*) and each proposer's best offer is the prescribed one (a rejected offer gives $\delta^2 s^* < s^*$). By the one-stage deviation principle the profile is subgame perfect, with immediate agreement on $(\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$.